AKSHAT PANKAJ VISHWAKARMA

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AI & Full-Stack Engineer and 3rd-year B.Tech CSE (AI & ML) student at Sanjivani University. Experienced in architecting deployed diagnostic ML platforms, RAG knowledge retrieval systems, and high-performance REST APIs using Python, FastAPI, React, and SQL. Proven rapid prototyping and competitive technical execution with a **Top 35 finish at SunHacK International 2025** and participation across 7+ national hackathons.

EDUCATION

Sanjivani University | *B.Tech in Computer Science & Engineering (AI & ML)* **Maharashtra, India | 2024 - 2028**

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Management Systems (SQL), Operating Systems, Object-Oriented Programming (OOP), RESTful API Architecture

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, JavaScript (ES6+), SQL, HTML5, CSS3, R
- **AI / ML & Data:** TensorFlow, Scikit-learn, OpenCV, Pinecone (Vector DB), RAG Pipelines, Prompt Engineering
- **Backend & DB:** FastAPI, Flask, REST APIs, PostgreSQL, MySQL, SQLite, Database Modeling, CRUD Architecture
- **Frontend & UI:** React, Tailwind CSS, Responsive Web Design, Figma (UI/UX, Wireframing, Component Systems)
- **Dev Tools & Cloud:** Git, GitHub, Docker, Postman, Vercel Deployment, VS Code, Power BI, Debugging & Rapid Prototyping

PROJECTS

GeneGuard 2.0 | *AI Genomic Health & Clinical RAG Platform* **[Live Demo] [Code]** **React, FastAPI, Scikit-learn, Pinecone, Vercel**

- Architected a full-stack diagnostic AI platform deploying 5 specialized ML pipelines (Cardiovascular, Metabolic, Thyroid), achieving **80.2% ROC-AUC** and **77.4% precision** on benchmark evaluations.
- Implemented an evidence-backed Medical RAG engine combining **Pinecone vector database** with *all-MiniLM-L6-v2* embeddings over the Gale Encyclopedia of Medicine for hallucination-resistant clinical Q&A.
- Developed an interactive multi-generational family risk tree isolating hereditary risk factors and built an OCR document processing pipeline for automated laboratory blood report extraction.
- Deployed decoupled architecture with sub-200ms API inference latency via FastAPI microservices and responsive React frontend.

SupplyGuard AI | *Predictive Logistics & Disruption Forecasting* **[Live Demo] [Code]** **Python, TensorFlow, PostgreSQL, React**

- Engineered an intelligent freight analytics system using **TensorFlow & Scikit-learn** to forecast global shipment bottlenecks, multi-echelon inventory risks, and port transit delays.
- Built dynamic route optimization and scenario-analysis workflows analyzing carrier telemetry, weather anomalies, and historical disruption patterns.
- Designed transactional PostgreSQL data schemas and rendered interactive real-time risk dashboards in React, reducing manual tracking overhead.

Smart College ERP System | *Institutional Management & Anti-Proxy Engine* **React, FastAPI, SQL, Geolocation, Dynamic QR**

- Developed an enterprise campus ERP system featuring an anti-proxy attendance verification engine combining **dynamically refreshed 15-second QR codes** with radius geolocation validation.
- Architected relational database schemas and automated workflows for student grading, timetable scheduling, and institutional reporting supporting 1,000+ prospective users.
- Built role-based access control (RBAC) dashboards with granular permission tiers for administrators, faculty, and students.

EXPERIENCE & HACKATHON ACHIEVEMENTS

Hackathons & Technical Competitions | *Lead Developer & Finalist* **National & International | 2025 - 2026**

- **Top 35 Finalist - SunHacK International 2025:** Pitched an AI-driven prototype selected among top entries globally at Sandip University under intensive 36-hour sprint conditions.
- **7+ National Hackathons:** Competed in **Smart India Hackathon (SIH) 2026**, **DIPEX 2025 & 2026**, and **Eureka 2025**; led cross-functional teams translating ambiguous problem statements into production-ready software prototypes.

Freelance UI/UX & Graphic Designer | *Freelance* **Remote | 2023 - Present**

- Designed responsive web interfaces, interactive prototypes, and design systems in Figma for commercial clients, ensuring WCAG accessibility and brand consistency.
- Partnered directly with founders and developers to iterate on user experience wireframes, delivering 20+ design assets with 100% on-time milestone delivery.

CERTIFICATIONS

- **IBM:** Python for Data Science, R Programming, IBM Watson AI, DevOps Foundations, SPSS Statistics
- **Great Learning:** Prompt Engineering Specialization, Data Science Foundation